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looked at first by our left eye L (closing the right eye) and then by our right eye R (closing the left eye), the position of the pen changes with respect to the back ground of the wall. *The shift in the position of an object (say, a pen) when viewed with two eyes, keeping one* eye closed at a *time is known as Parallax*. The distance between the left eye (L) and the right eye (R) in this case is the basis.

$\angle$LOR is called the *parallax angle or parallactic angle*.

Taking LR as an arc of length b and radius LO = RO = $x$

we get $\theta = \frac{b}{x}$; b-basis, $x$-unknown distance

Knowing 'b' and measuring θ, we can calculate $x$.

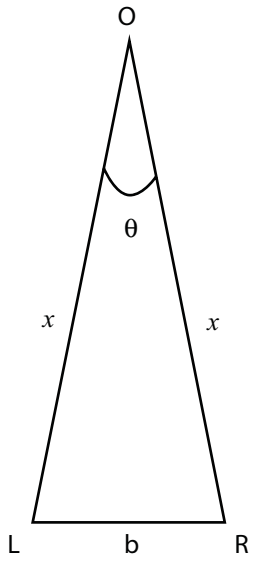



**Figure 1.4** Parallax method

**Determination of distance of Moon from Earth**

In Figure 1.5, C is the centre of the Earth. A and B are two diametrically opposite places on the surface of the Earth. From A and B, the parallaxes $\theta_1$ and $\theta_2$ respectively of Moon M with respect to some distant star are determined with the help of an astronomical telescope. Thus, the total parallax of the Moon subtended on Earth $\angle AMB = \theta_1 + \theta_2 = \theta$.

If θ is measured in radians, then $\theta = \frac{AB}{AM}$; AM≈MC

$\theta = \frac{AB}{MC}$ or $MC = \frac{AB}{\theta}$; Knowing the values of AB and θ,

we can calculate the distance MC of Moon from the Earth.

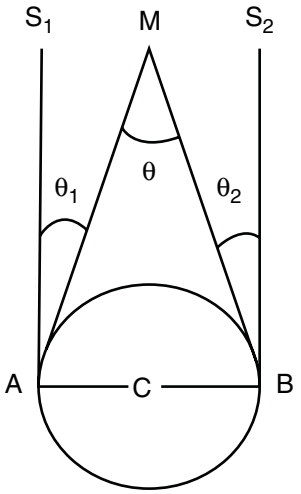



**Figure 1.5** Parallax method determination of distance of Moon from Earth

## EXAMPLE 1.2

The Moon subtends an angle of 1° 55′ at the base line equal to the diameter of the Earth. What is the distance of the Moon from the Earth? (Radius of the Earth is $6.4 \times 10^6 m$)